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Dietary supplementation with Zyflamend poly-herbal extracts and fish oil inhibits intimal hyperplasia development following vascular intervention

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ABSTRACT

The polyherbal blend Zyflamend™ has been shown to have anti-inflammatory properties and attenuate inflammatory-modulated pathologies. Fish oils have also been shown to have cardioprotective properties. However, the beneficial effects of their combination have not been investigated. Intimal hyperplasia (IH), a pathological remodeling response of a vessel to injury, is heavily regulated by an immune-mediated reaction. The objective of this study was to determine if dietary supplementation with Zyflamend and/or Wholemega could affect inflammatory-dependent vascular remodeling mechanisms when provided at human equivalent doses. Based on their anti-inflammatory properties and protective benefits demonstrated in previous pre-clinical studies, we hypothesized administration of these supplements would prevent IH in an animal model of vascular injury. The diets of aged male rats were supplemented with human equivalent doses of Zyflamend (Zyf) and/or Wholemega (WMega) or placebo (Plac) for 1wk prior to balloon angioplasty (BA)-induced injury of the left carotid artery. At 28d post-injury morphometric analysis of carotid tissue revealed IH was decreased in Zyf + WMega animals compared to placebo, while Zyf or WMega independently had no significant effect. Serum cytokine screening indicated injury-induced interleukin family isoforms, interferon-γ, and macrophage inflammatory proteins were downregulated by Zyf + WMega. Immunohistochemical staining for monocyte/macrophage phenotypic markers revealed that while overall monocyte/macrophage vessel infiltration was not affected, Zyf + WMega limited the alternative differentiation of M2 macrophages and reduced the presence of myofibroblasts in the injured vessel wall. In

Abbreviations: IH, intimal hyperplasia; Zyf, Zyflamend; WMega, Wholemega; Plac, placebo; BA, balloon angioplasty; VSMC, vascular smooth muscle cell; EPA, eicosapentaenoic acid; DHA, docosahexaenoic acid; PUFA, polyunsaturated fatty acids; LA, linoleic acid; ALA, alpha-linoleic acid; I:M, intima:media ratio; MMP, matrix metalloproteinase; TIMP, tissue inhibitor of MMP; PCNA, proliferating cellular nuclear antigen; MAC387, macrophage+monocyte antigen; MSR, macrophage scavenger receptor; Vim, vimentin; SMA, smooth muscle α-actin; IL-12p70, interleukin-12; IFNg, interferon-γ; IP-10, interferon-γ induced protein-10; MIP-2, macrophage inflammatory protein-2; CRP, c-reactive protein; NFκB, nuclear factor κB; TNFα, tumor necrosis factor α; COX2, cyclooxygenase 2; M2, type 2 macrophage.

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summary, dietary supplementation with Zyf + WMega attenuated the acute inflammatory response following vascular injury and inhibited IH development in vivo.

1. Introduction

Dietary herbal supplements have been gaining in popularity for the prevention and improvement of various disease processes, including vascular disease. There is evidence that the combinatorial use of multiple herbal supplements can act synergistically to potentiate their individual efficacy [1,2]. Zyflamend is a poly-herbal dietary supplement composed of the extracts of holy basil, turmeric, ginger, green tea, rosemary, Hu Zhang, barberry, oregano, baikal skullcap, and Chinese goldthread. Wholemega is an extra-virgin whole fish oil preparation derived from wild-caught Alaskan salmon, composed of a variety of polyunsaturated fatty acid constituents. Experimental studies have shown Zyflamend is a potent anti-inflammatory modulator and is protective in a variety of cellular processes involved in inflammatory-modulated pathogenesis such as tumor progression and osteoclastogenesis [3-9]. Experimental evidence has shown that omega-3 fatty acids can be protective against pathological vascular remodeling and vascular disease progression [10-12], and the American Heart Association has recommended the use of dietary supplements in moderation in order to reduce the risk of cardiovascular-related events.

Pro-inflammatory processes heavily regulate vascular disease progression. Furthermore, vascular intervention failures stem from the acute inflammatory response to mechanical injury, often requiring additional intervention procedures with increased morbidity. Intimal hyperplasia (IH)-induced restenosis is the characteristic consequential response of a vessel to injury, where phenotypic modification of medial vascular smooth muscle cells (VSMC) and their intimal migration has been accepted as the basis of lesion formation for decades. More recently, additional evidence has shown that monocyte/macrophage infiltration is an independent predictor of restenosis and can contribute to IH development by serving as precursors to neointimal myofibroblasts [13]. Previous studies from our laboratory have focused on defining the role of inflammatory mechanisms and matrix remodeling enzymes in IH development [14-19]. A potent anti-inflammatory agent could prove beneficial in diminishing or preventing dysfunctional vascular remodeling and IH pathogenesis by attenuating these pro-inflammatory mechanisms.

The objective of this study was to determine if dietary supplementation with Zyflamend (Zyf) and/or Wholemega (WMega) could affect inflammatory-dependent vascular remodeling mechanisms. Here, we administered both supplements at human equivalent doses in experimental rodent diets, examined their effects on inflammatory family profiles as a means of delineating potential downstream remodeling effectors, and assayed established vascular remodeling markers and overall vessel wall thickening following balloon angioplasty induced injury. Based on the protective benefits demonstrated in previous pre-clinical studies of inflammatory-modulated pathologies, we hypothesized that administration of these supplements would prevent IH development.

2. Methods and materials

2.1. Supplements

Zyflamend™ (Zyf; New Chapter, Brattleboro, VT) is a dietary supplement derived from the extracts of ten different herbs. A detailed description and characterization of the preparation of Zyf and quality assurance of the mixture has been described [3]. Wholemega™ (New Chapter, Brattleboro, VT) is composed of extra-virgin wild Alaskan salmon oil, extracted using a mechanical separation process similar to that of extra virgin olive oil, combined with supercritical extracts of rosemary and oregano to provide antioxidant stability. Eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA) comprise 20% (w/w) of the oil.

2.2. Diets

The background diet used here was based on the composition of a typical Western diet to maximize translatability, with characteristic energy distributions (50% carbohydrate, 34% fat, 16% protein) and lipid compositions [~15% monounsaturated, ~13% saturated, and 6%–7% polyunsaturated fatty acids (PUFA) where the PUFA distribution included 6% linoleic acid (LA) and 0.7% alpha-linoleic acid (ALA)]. Human equivalent doses of these PUFA for the rodent diets were derived from human daily intakes of 15 g and 1.33 g for LA and ALA, respectively, matching human intakes for n-6 and n-3 PUFA as described previously [20,21].

Experimental diets were formulated on the background diet described above, and the composition of placebo (Plac); Zyflamend+Wholemega (Zyf + WMega), Zyflamend (Zyf), and Wholemega (WMega) diets are presented in Table 1. The level of poly-herbal extracts from Zyf in the rodent diet was 0.21% (g/kg), allometrically scaled to a human equivalent dose of three 1 g capsules [21]. The level of WMega in the diet was equivalent to 1.27% energy, or a human equivalent dose of 2.8 g/d (0.56 g/d of EPA + DHA). Extrapolations were chosen using metabolic activity based on data evaluating interspecies comparisons with energy and non-energy producing nutrients [20-22]. Rodent diets were custom produced by Research Diets, Inc. (New Brunswick, NJ). The product numbers for the specific diets are #D07100506 for placebo, #D07100507 for Zyflamend+Wholemega, D07100508 for Zyflamend, and #D0700509 for Wholemega.

2.3. Animals

Male Sprague–Dawley rats (8–10 months of age) were obtained from Harlan Laboratories, Inc. (Indianapolis, IN). All animal

Table 1 – Experimental Diet Compositions

	Placebo		Zyf + WMega		Zyf		WMega	
Ingredients	g/kg	kcal	g/kg	kcal	g/kg	kcal	g/kg	kcal
Casein, Lactic	169.5	678.0	169.5	678.0	169.5	678.0	169.5	169.5
L-Cystine	3.4	13.6	3.4	13.6	3.4	13.6	3.4	3.4
Corn starch	342.4	1369.6	342.4	1369.6	342.4	1369.6	342.4	342.4
Maltodextrin 10	84.8	339.0	84.8	339.0	84.8	339.0	84.8	84.8
Sucrose	113.0	452.0	113.0	452.0	113.0	452.0	113.0	113.0
Cellulose	56.5	0.0	56.5	0.0	56.5	0.0	56.5	56.5
Cocoa butter, deodorized	38.4	345.8	38.4	345.8	38.4	345.8	38.4	38.4
Linseed oil, RBD	4.6	41.8	4.6	41.8	4.6	41.8	4.6	4.6
Palm oil, bleached, deodorized	53.9	484.8	53.9	484.8	53.9	484.8	53.9	53.9
Safflower oil, USP	29.3	263.3	29.3	263.3	29.3	263.3	29.3	29.3
Sunflower oil, Trisun Extra	27.8	249.7	27.8	249.7	27.8	249.7	27.8	27.8
Zyflamend	0.0	0.0	6.0	32.8	6.0	32.8	0.0	0.0
Extra virgin olive oil	12.0	107.4	0.0	0.0	6.0	53.7	6.0	12.0
Wholemega	0.0	0.0	6.0	54.2	0.0	0.0	6.0	0.0
t-BHQ	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Mineral Mix S10026	11.3	0.0	11.3	0.0	11.3	0.0	11.3	11.3
Calcium phosphate, dibasic	14.7	0.0	14.7	0.0	14.7	0.0	14.7	14.7
Calcium carbonate	6.2	0.0	6.2	0.0	6.2	0.0	6.2	6.2
Potassium citrate, 1 H2O	18.6	0.0	18.6	0.0	18.6	0.0	18.6	18.6
Vitamin mix V13401	11.3	45.2	11.3	45.2	11.3	45.2	11.3	11.3
Choline bitartrate	2.3	0.0	2.3	0.0	2.3	0.0	2.3	2.3
alpha vitamin E acetate 500 IU/gm	0.1	0.0	0.1	0.0	0.1	0.0	0.1	0.1
Total	**1000**	**4391**	**1000**	**4370**	**1000**	**4391**	**1000**	**1000**

Macronutrient content in each of the diets were identical: Protein, carbohydrate and fat content (g/kg diet) were 19.2 (16% kcal), 62.2 (50% kcal) and 19.2 (34% kcal), respectively. Each diet contained 4.4 kcal/g. Macronutrient content in each of the diets were identical: Protein, carbohydrate and fat content (g/kg diet) were 19.2 (16% kcal), 62.2 (50% kcal) and 19.2 (34% kcal), respectively. Each diet contained 4.4 kcal/g. In the diets containing Wholemega (Wmega), the amounts of EPA and DHA were 0.54 g/kg diet and 0.66 g/kg diet, respectively. When extrapolated, the daily human equivalent doses of eicosapentaenoic acid (EPA) and docosapentaenoic acid (DHA) in these diets translate to 250 mg/d (0.11% energy) and 310 mg/d (0.14% energy), respectively [21]. Abbreviations: Zyf, Zyflamend; WMega, Wholemega; Zyf + WMega, Zyflamend + Wholemega.

procedures conformed to the Guide for the Care and Use of Laboratory Animals published by the National Institutes of Health (NIH Publication No. 85–23, Revised 1996) and were approved by the Institutional Animal Care and Use Committee, certified by the American Association of Accreditation of Laboratory Animal Care. Animals were provided free access to fresh food and water, and were housed in individual cages in order to monitor daily food/supplement intake. All animals were placed on the Western-equivalent placebo diet for 1wk for acclimation, and then randomly assigned to supplement groups and provided either Plac, Zyf + WMega, Zyf, or WMega supplemented diets continuously thereafter. Experimental diets were replaced every other day, and individual animal consumption was recorded. Individual effects were normalized to each animal's mean daily intake, prior to cohort data analytics, in order to control for variations in ad libitum dose exposure. Body weight changes were recorded over the duration of the study as a method of animal health analyses and surgical procedure recovery monitoring. Body weights were recorded at baseline (at study initiation and start of Western-equivalent placebo diet), at experimental supplement diet switch (1 week post placebo acclamation), at the time of BA (1 week of experimental diet supplementation), and at 7 d and 14 d post-surgery.

2.4. Balloon angioplasty vascular injury and tissue procurement

After 1wk of experimental diet supplementation, all animals underwent vascular injury via balloon angioplasty (BA) of the left common carotid artery using a standard and accepted pre-clinical model of vascular disease development and progression [15,17]. Briefly, rats were anesthetized with vaporized 1–5% isofluorane. A midline neck incision exposed the left carotid artery. A 2F balloon catheter was passed through a small arteriotomy in the external carotid artery, into the common carotid artery, and down to the aortic arch. The balloon was distended to 2 atm of pressure and passed anti-grade and retro-grade through the common carotid to denude the endothelium. This was repeated three times, after which the catheter was removed, the external carotid ligated, and the incision closed. The right carotid artery was exposed by partial dissection and served as the non-injured operational sham to confirm no preexisting vascular disease in any experimental animal. At 28d post-injury animals were euthanized with an overdose of CO_2 gas and the thoracic cavity opened for full exposure. Vascular perfusion and fixation was performed through a 16-gauge needle inserted into the left ventricle and attached to a saline bag pressure controlled identically by gravity across all procurement replicates. A small incision was made into the right atrium for fluid draining. After 10 min of saline perfusion the bag was switched to fresh 10% formalin for 10 min for in situ tissue fixation, and the carotid arteries were extracted by fine micro-dissection [15,17].

2.5. *Inflammatory cytokine immunoassay*

Serum was collected from each animal at the time of vascular injury (baseline), 7d post-injury, and 28d post-injury, and frozen at 80 °C until analyses. Cytokine Rat Membrane Antibody Arrays (Abcam, Cambridge, MA) were used to qualitatively screen 34 cytokine serum antigens. Serum aliquots (25 mL) from each 28d experimental sample (n = 9–10 surviving 29d per group) were pooled together according to supplement group and ran in batch on independent membranes, according to the manufacturer's instructions. Briefly, capture antibodies were spotted on nitrocellulose membranes, and pooled serum samples from each group were incubated overnight on separate arrays for antigen capture. Following capture, a supplied cocktail of biotin-conjugated anti-cytokine antibodies was used for secondary binding, labeling was performed with streptavidin HRP, and immune complexes were detected using chemiluminescence and autoradiography for qualitative comparison. Milliplex Rat Cytokine and Chemokine Magnetic Bead Panel (EMD Millipore Corp, Billerica, MA) was used to quantitatively measure 27 cytokine serum antigens using the Luminex 200 multiplexing immunoassay system (Life Technologies Corp, Carlsbad, CA). Serum aliquots from each baseline, 7d, and 28d sample were analyzed independently in duplicate, according to the manufacturer's instructions. Because sample number was limited by the supplied aliquots of magnetic beads, in order assay samples at all stated time point (baseline, acute inflammation at 7d post-BA, and chronic inflammation at 28d post-BA) the number of experimental samples possible for this analysis was limited, and multiplex arrays were performed on subpopulation of each study group at n = 6. The animal cohort chosen for this assay was randomly assigned, and not selected based upon bias of any other outcome of the study. Briefly, independent samples were mixed with supplied magnetic beads, pre-coated with cytokine-specific antibodies, and color coded for antigen specificity. Following antigen capture, a supplied cocktail of biotin-conjugated cytokine-specific antibodies was used for secondary binding, labeling was performed with streptavidin-PE, and labeled beads were magnetically captured for quantification. Using Luminex 200 multiplexing automation, captured bead color was electronically decoded for antigen identification and PE signal was simultaneously fit to supplied standards for antigen quantification.

2.6. *Morphometric analysis and immunohistochemistry*

Carotid arteries were placed in 10% formalin solution for at least 24 h post procurement for further fixation, then paraffin embedded, and cross sectioned transversely at 4 μm thickness. Tissue sections were stained with Masson's Trichrome elastin for morphometric analysis, according to standard operating procedures. Images were acquired at 100× magnification using a BX51 Olympus microscope with an Olympus Q-color camera. Intimal, medial, and luminal areas were measured in μm^2 in four independent segments of each carotid sample (one proximal, mid-proximal, mid-distal, and distal segment of each sample), using Image ProPlus image analysis software, and the mean ratio of intimal and medial areas (I:M), normalized to the animal's mean daily intake, was used as a quantification of IH development in each sample. Normalized individual sample means were then averaged for all replicates in each study cohort, and the data are presented as means ± SEM for each supplement group.

Additionally, sections were prepared for immunohistochemical analyses for tissue level protein or antigenic marker quantification. Sections were deparaffinized and stained with isoform-specific matrix metalloproteinase (MMP)-9 (Dako Corp, Carpentaria, CA, USA), MMP-2, MMP-14, tissue inhibitor of MMP (TIMP)-1, TIMP-2 (Millipore Corp, Billerica, MA, USA), proliferating cellular nuclear antigen (PCNA), macrophage+monocyte antigen (MAC387), or macrophage scavenger receptor (MSR; Abcam, Cambridge, MA, USA) primary antibodies, according to manufacturers' instructions. Detection was carried out using a DAB-substrate detection kit (InnoGenex, San Ramon, CA, USA) or DyLight fluorescently conjugated secondary antibodies (Abcam). Co-staining for Vimentin (Vim; Invitrogen Corp, Carlsbad, CA, USA) and smooth muscle α-actin (SMA; Dako Corp) was performed simultaneously with primary antibodies and differentially-conjugated, species-specific secondary antibody cocktails. Using the same acquisition equipment and analysis software, images were acquired at 100× and 400× magnification. Intimal or whole vessel staining of each antigen of interest was quantified at 400× by measuring the % area present as brown or fluorescent pixels, in independent duplicates of one mid-proximal and one mid-distal segment of each sample. Vim + SMA co-localization staining was quantified using 400× image overlays in independent duplicates of one mid-proximal and one mid-distal segment of each sample. The mean % area stained was used as a quantification of tissue level protein expression or as a quantification of cell-type specific antigenic markers in each sample. Individual sample means were then averaged for all replicates in each study cohort, and the data are presented as means ± SEM for each supplement group.

Table 2 – Vascular areas

	Placebo	Zyf + WMega	Zyf	WMega
Intima	0.112 ± 0.015	0.066 ± 0.011	0.105 ± 0.012	0.105 ± 0.011
Media	0.138 ± 0.015	0.119 ± 0.011	0.101 ± 0.005	0.109 ± 0.008
Lumen	0.247 ± 0.033	0.228 ± 0.016	0.215 ± 0.016	0.222 ± 0.017

Vascular layer areas measured as μm^2. Values are means ± SEM. Abbreviations: Zyf, Zyflamend; WMega, Wholemega; Zyf + WMega, Zyflamend +Wholemega.

2.7. Statistical analyses

All data are reported as means ± SEM. Statistical analyses were performed using Student t-tests or one-way ANOVA and post hoc Student–Newman–Keuls tests using SigmaStat 3.5 software (Systat Software Inc., Chicago, IL). All assays were powered to ≥0.8 based on a priori sample calculation. Probability (*P*) values ≪0.05 were considered significant.

3. Results

3.1. I:M was decreased in Zyf + WMega supplemented animals compared to placebo controls, while Zyf or WMega alone had no significant effects

At 28d post injury intimal, medial, and luminal areas were measured in μm^2 as described (Table 2), and I:M ratios were used to quantify the degree of IH development in response to injury. The operational shams of all animals used for this study were microscopically examined and exhibited no measureable intimal layer and no preexisting or under-lying vascular pathology. Marked intimal thickening was present in the BA samples of animals receiving the placebo diet (I:M = 0.82 ± 0.12, Fig. 1A and B), while I:M was significantly decreased in animals receiving Zyf + WMega supplementation (I:M = 0.43 ± 0.14; *P ≪ .05 vs placebo, n = 10; Fig. 1A and B). Supplementation with Zyf or WMega independently had no statistically significant effect on intimal thickening compared to placebo (I:M = 0.76 ± 0.07 and 0.71 ± 0.04, respectively, Fig. 1A and B).

3.2. Experimental diets did not differentially affect changes in rodent body weight

Rodent weights were recorded at baseline (at study initiation and start of Western-equivalent placebo diet), at experimental supplement diet switch (1 week post placebo acclamation), at the time of BA (1 week experimental diet consumption), and at 7d and 14d post-BA as a method of animal health analyses and surgical procedure recovery monitoring. Trends in body weight change over time were analogous in all study groups (Fig. 2), with animals in all group gaining ~30-35 g up until the time of BA injury, after which weights dropped slightly but relatively equivalently in all groups, as is expected during post procedural recovery. Furthermore, average daily consumption over the course of the study was not significantly different among groups. Placebo control animals consumed an average of 17.4 ± 0.7 g/day, while Zyf + WMega, Zyf, and WMega animals consumed 16.2 ± 0.5, 15.0 ± 0.4, and 15.1 ± 0.5 g/day, respectively (*P*=NS, n = 9–10). These individual consumption rates were directly related to baseline body weight at the initiation of the study (i.e. animal cohorts with a lower mean weight at the beginning of the study demonstrated overall lower average consumption).

3.3. Injury-induced inflammatory cytokines were decreased systemically in Zyf + WMega supplemented animals compared to placebo controls

Gross qualitative analyses via cytokine screening antibody arrays indicated interleukin family isoforms and macrophage inflammatory proteins were decreased at 28d in the

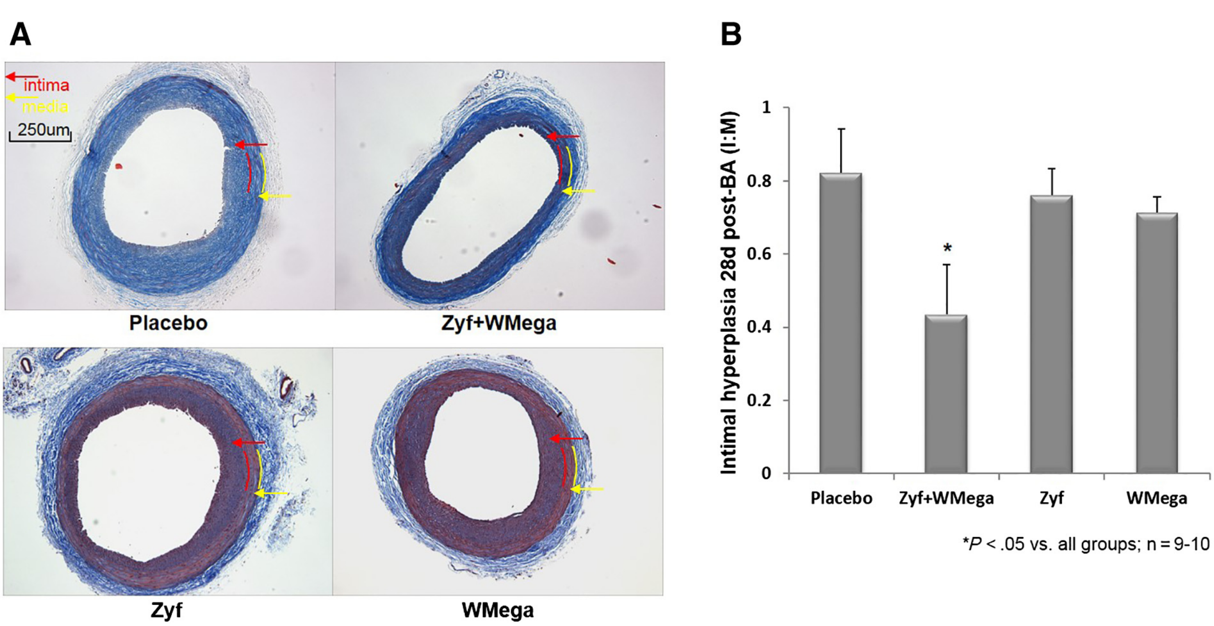

Fig. 1 – Zyf + WMega supplementation decreases I:M. At 28d post-BA, carotid artery sections were procured and stained with Masson's Trichrome elastin for measurement of intima area vs media area ratio (I:M) to quantify degree of IH development. A. Images, taken at 100× magnification, are representative of post-BA injury samples of each experimental group. Abbreviations: Zyf, Zyflamend; WMega, Wholemega; Zyf + WMega, Zyflamend +Wholemega. Internal elastic lamina (IEL) in each image are marked by red arrows and external elastic lamina (EEL) are marked by yellow arrows. B. Graph exhibits means ± SEM, n = 9–10. *P<0.05 vs all other groups.

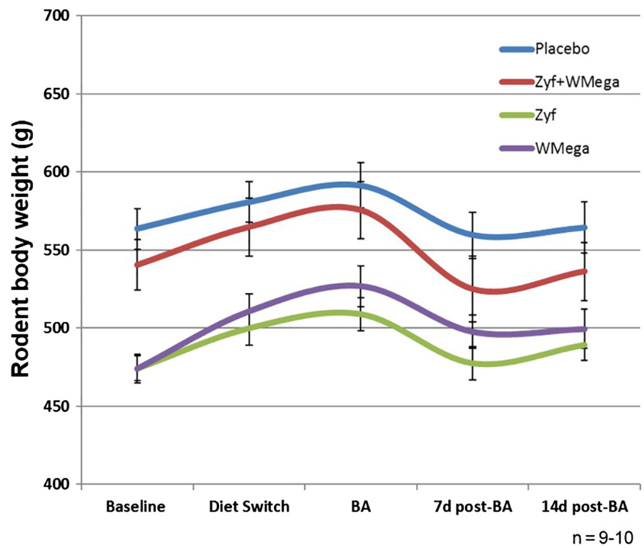

Fig. 2 – Experimental diets did not differentially affect changes in body weight. Changes in the body weight of each rodent were recorded over the duration of the study as a method of animal health analyses and surgical procedure recovery monitoring. Body weights were recorded in grams (g) at baseline (at study initiation and start of Western-equivalent placebo diet), at experimental supplement diet switch (1 week post placebo acclamation), at the time of BA (1 week experimental diet consumption), and at 7d and 14d post-BA. Abbreviations: Zyf, Zyflamend; WMega, Wholemega; Zyf + WMega, Zyflamend +Wholemega. Graph exhibits means ± SEM, n = 9–10.

serum of Zyf + WMega supplemented animals compared to placebo (Fig. 3). Quantitative analyses via multiplex analysis indicated Zyf + WMega supplementation for 1wk prior to injury had no effect on the baseline inflammatory status of study animals compared to placebo controls. However, Zyf + WMega did significantly decrease multiple injury-induced inflammatory mediators in the acute response post-injury Specifically, leptin, interleukin-12 (IL-12p70), interferon-γ (IFNg), and macrophage chemokines interferon gamma-induced protein-10 (IP-10) and macrophage inflammatory protein-2 (MIP-2) were markedly downregulated in the serum of supplemented animals at 28d post-BA (*P ≪ .05 vs placebo, n = 6; Table 3).

3.4. *Dietary supplementation had no significant effect on matrix remodeling or proliferative cell markers*

Injury-induced intimal expressions of MMP-mediated matrix remodeling mechanisms were not significantly different in Zyf + WMega supplemented animals, nor Zyf or WMega alone, compared to animals receiving the placebo control diet (Table 4). Likewise, intimal PCNA levels were not different between the two groups (Table 4). For further analyses, all assayed matrix enzymes and enzyme activators (MMP-2, MMP-9, MMP-14) were pooled as a measure of pro-modeling isoforms, and all enzyme inhibitors (TIMP-1, TIMP-2) were pooled as a measure of anti-remodeling isoforms. When pooled data sets were normalized to placebo as a measure of % change, Zyf + WMega combination slightly down-regulates pro-remolding markers, though not to a significant degree (n = 28–30; Fig. 4).

3.5. *MSR presence and the co-localization of vim + SMA cell markers were significantly decreased in the arterial tissue of Zyf + WMega supplemented animals compared to placebo controls*

The presence of MSR was significantly decreased in the injured carotids of Zyf + WMega supplemented animals compared to placebo controls (*P ≪ .01, n = 10; Fig. 5A and B, Table 4). Supplementation with Zyf or WMega independently had no significant effect on the presence of MSR markers. Likewise, the presence of macrophage+monocyte antigen MAC387 was not affected in the tissue of any group (n = 10; Fig. 5B, Table 4). Tissue co-localization of Vim + SMA cell markers was significantly decreased in the tissue of Zyf + WMega animals vs placebo controls (% vessel wall co-stained = 23.1 ± 1.9 vs 32.1 ± 3.3; *P ≪ .05 vs placebo, n = 9–10; Fig. 6A and B), while Zyf or WMega alone had no effect.

4. Discussion

A heart-healthy diet, low in fat and high in fiber and omega-3 fatty acids, is a hallmark recommendation for risk management of vascular disease. However, the American Heart Association has acknowledged that diet alone may not be sufficient to reach a protective level of intake for many dietary nutrients, and has recommended physician-managed use of dietary supplements as adjuvant therapy for primary and secondary prevention of cardiovascular disease [10].

There is evidence that the use of herbal botanicals and phytochemicals in combination can act synergistically to

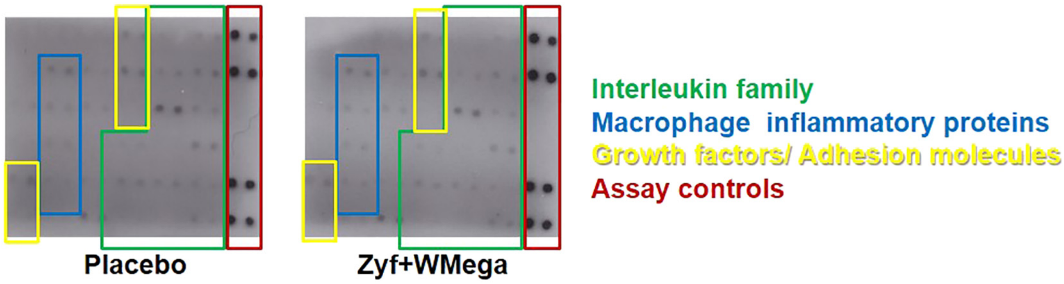

Fig. 3 – Zyf + WMega supplementation systemically decreases injury-induced inflammatory cytokines. Qualitative inflammatory antibody arrays were used to screen 34 cytokine antigens in rodent serum. Serum aliquots (25 μl) from each experimental sample taken at 28d post-BA were pooled together by treatment group and ran in batch on independent membranes. Panel image is representative of autoradiographic signal of each experimental group. Abbreviation Zyf + WMega, Zyflamend+Wholemega. Outlines are used to highlight assay controls and separate cytokine families of interest.

potentiate their efficacy when compared to individual components alone [1,2]. In the current study the combination of Zyf + WMega demonstrates the integration of a natural herbal botanical complex with food grade wild Alaskan salmon oil that retains its full native complement of polyunsaturated fatty acids. It is well established that fish oil and omega-3 PUFAs are protective against dysfunctional vascular remodeling and vascular pathogenesis in experimental models of disease [11,12]. There is some supportive clinical evidence showing a reduction in triglycerides, hypertension, and hypercoagulability [23,24]. Zyf is an anti-inflammatory agent that has been shown to attenuate inflammatory-modulated pathologies such as osteoclastogenesis and tumor progression in animal models of disease [3-5,8,9]. However, prior to our study, Zyf has not been investigated in the prevention of vascular disease progression or its pro-inflammatory modulated pathways. Here we demonstrated that dietary supplementation with human equivalent doses of Zyf + WMega significantly inhibited IH development in vivo in an animal model of vascular injury. We also show that this attenuation is an apparent result of the interactive effects of the constituents to potentiate protection beyond what would be provided by Zyf-derived herbal botanicals or fish oil administration alone. Furthermore, by normalizing vessel thickening to the degree of dietary supplement intake and demonstrating equivalent daily food intake and equivalent body weight changes over time, we determine these protective effects are a direct results of attenuated vascular remodeling, not an experimental artifact seen as a result of differential food intake or animal size discrepancies. These

Table 3 – Quantitative cytokine serum analysis

	At injury		7d post-BA		28d post-BA	
	Placebo	Zyf + WMega	Placebo	Zyf + WMega	Placebo	Zyf + WMega
Macrophage inflammatory protein-1a	13.9 ± 1.4	13.0 ± 0.8	20.6 ± 2.6	21.3 ± 2.7	28.5 ± 5.6	28.2 ± 5.3
Macrophage inflammatory protein-2	104.2 ± 6.0	45.3 ± 1.4	76.2 ± 11.4	103.9 ± 11.9	161.3 ± 36.3	95.9 ± 7.2 *
Monocyte chemoattractant protein-1 [n]	1.1 ± 0.1	1.0 ± 0.1	1.3 ± 0.1	1.2 ± 0.1	1.7 ± 0.2	1.2 ± 0.4
Interleukin-1b	———	———	13.6 ± 4.9	15.5 ± 2.2	44.2 ± 16.5	62.8 ± 1.8
Interleukin-2	24.6 ± 1.8	23.9 ± 7.6	44.0 ± 9.8	36.9 ± 12.4	71.7 ± 16.7	84.7 ± 24.5
Interleukin-17A	24.6 ± 2.2	16.6 ± 6.9	35.1 ± 8.5	31.8 ± 2.5	42.4 ± 11.5	31.0 ± 11.8
Interleukin-10	36.5 ± 4.1	25.9 ± 5.6	39.1 ± 2.8	41.3 ± 13.4	71.1 ± 15.1	41.7 ± 11.3
Interleukin-12p70	173.6 ± 23.4	115.0 ± 36.4	286.1 ± 32.5	208.0 ± 34.6	244.9 ± 5.3	196.3 ± 6.6 *
Interleukin-5	256.6 ± 14.0	234.0 ± 24.5	304.2 ± 24.2	254.5 ± 8.4	211.2 ± 47.2	253.4 ± 29.1
Interleukin-18	121.9 ± 34.8	116.2 ± 10.3	152.7 ± 15.3	147.2 ± 33.6	174.7 ± 33.9	142.8 ± 27.0
Interferon gamma (IFNg)	62.0 ± 14.2	115.0 ± 36.4	42.4 ± 13.5	208.0 ± 34.5	362.9 ± 35.6	127.3 ± 18.6 *
IFNg induced protein 10	309.0 ± 17.9	254.7 ± 13.9	275.6 ± 28.7	327.1 ± 34.1	553.6 ± 12.5	325.4 ± 20.6 *
Leptin [n]	33.5 ± 5.3	27.2 ± 5.8	14.0 ± 2.5	15.4 ± 1.2	20.1 ± 4.9	8.1 ± 2.4 *
Growth-regulated oncogene chemokine	58.1 ± 6.9	64.3 ± 2.5	57.9 ± 10.6	76.4 ± 10.2	178.6 ± 58.4	83.3 ± 51.6
Vascular endothelial growth factor	44.4 ± 1.9	31.4 ± 3.2	43.3 ± 2.2	50.3 ± 1.4	77.5 ± 19.6	47.7 ± 14.2
Fractalkine	61.7 ± 4.4	53.3 ± 2.5	52.6 ± 2.5	101.1 ± 2.5	276.6 ± 100.9	99.2 ± 92.1
LIX [n]	4.7 ± 0.3	3.9 ± 0.3	4.2 ± 0.2	4.5 ± 0.4	4.8 ± 0.1	4.4 ± 0.5
Epidermal growth factor	82.9 ± 41.4	98.0 ± 88.5	127.7 ± 66.2	205.9 ± 120.4	37.5 ± 10.5	86.3 ± 31.3
Tumor necrosis factor α	18.8 ± 1.7	16.9 ± 4.7	24.1 ± 1.1	19.1 ± 2.8	19.6 ± 4.9	19.2 ± 1.9
RANTES[n]	1.4 ± 0.1	1.4 ± 0.2	1.5 ± 0.1	1.5 ± 0.2	1.8 ± 0.2	1.5 ± 0.3

Serum cytokine concentrations represented as pg/ml except where noted [n] for ng/ml. Values are means ± SEM. *$P \ll .05$ vs Placebo cytokine level at equivalent time point, n = 6. Abbreviations: Zyf + WMega, Zyflamend +Wholemega; BA, balloon angioplasty.

Table 4 – Immunohistochemical analysis

	Placebo	Zyf + WMega	Zyf	WMega
MMP-2	20.2 ± 3.5	15.6 ± 1.1	20.1 ± 2.1	23.7 ± 1.6
MMP-9	17.0 ± 2.3	11.3 ± 1.9	19.9 ± 2.6	20.8 ± 2.4
MMP-14	8.3 ± 2.4	6.8 ± 0.9	5.0 ± 1.3	6.3 ± 1.5
TIMP-1	4.8 ± 0.8	5.1 ± 0.7	7.0 ± 1.7	5.4 ± 1.5
TIMP-2	13.3 ± 3.2	13.7 ± 3.1	7.5 ± 1.0	9.5 ± 2.1
PCNA	4.1 ± 0.9	2.6 ± 0.5	3.3 ± 0.5	4.8 ± 0.8
MAC387	18.0 ± 3.2	18.1 ± 1.7	16.4 ± 1.5	16.4 ± 2.0
MSR	8.5 ± 0.7	4.0 ± 0.4 *	9.4 ± 0.9	8.5 ± 0.7

Immunohistochemical staining represented as % vessel area stained. Values are means ± SEM. *$P \ll .01$ vs Placebo; n = 10. Abbreviations: Zyf, Zyflamend; WMega, Wholemega; Zyf + WMega, Zyflamend +Wholemega; MMP, matrix metalloproteinase; TIMP, tissue inhibitor MMP; PCNA, proliferating cellular nuclear antigen; MAC387, monocyte/macrophage antigen; MSR, macrophage scavenger receptor.

results suggest that this unique Zyf + WMega combination may be beneficial in vascular protection.

It is well established that peripheral vascular disease and atherosclerosis are pro-inflammatory modulated disease processes. Insult to the endothelial lining of the vessel leads to a disruption in the homeostasis of the vessel wall, whether that insult is induced by atherosclerosis or by mechanical injury during intervention. This results in the propagation of multiple inflammatory pathways, and ultimately the upregulation of cell adhesion molecules, pro-thrombotic elements, and growth factors [25]. It has been reported that C-reactive protein (CRP) can be used as a biomarker to predict atherosclerosis [26], and we have established the role of interleukin signaling in hormone-modulated IH development [14,19]. Zyflamend has been shown to attenuate inflammatory signaling via CRP, nuclear factor κB (NFκB), tumor necrosis factor α (TNFα), and cyclooxygenase 2 (COX2) [4,5]. We demonstrated that Zyf + WMega supplementation had no effect on the baseline inflammatory profile of our study animals, but did significantly decrease multiple injury-induced inflammatory mediators. Leptin, a cytokine whose elevation is associated with multiple cardiovascular related pathologies [27], was markedly decreased in supplemented animals compared to placebo controls during the chronic inflammatory state (28d post-injury), while no effect was demonstrated during injury-induced acute inflammation (7d post-injury). Interleukin family signaling was also affected by Zyf-WMega supplementation compared to Placebo, with the slight downregulation of active IL-12 homodimer IL-12p70 during acute inflammation and a significant reduction during the chronic inflammatory state. IL-12p70 is responsible for the activation of TNFα and IFNg, and can exert anti-angiogenic effects via IFNg-stimulated activation of macrophage chemokine IP-10 [28]. While IFNg and IP-10 were also downregulated during chronic inflammation in supplemented animals, Zyf + WMega did not have a significant effect on PCNA tissue level expression. This suggests that the anti-inflammatory effects demonstrated here exert a mechanism of protection independent of controlling dysfunctional VSMC proliferation and points to a potential for long-term macrophage involvement.

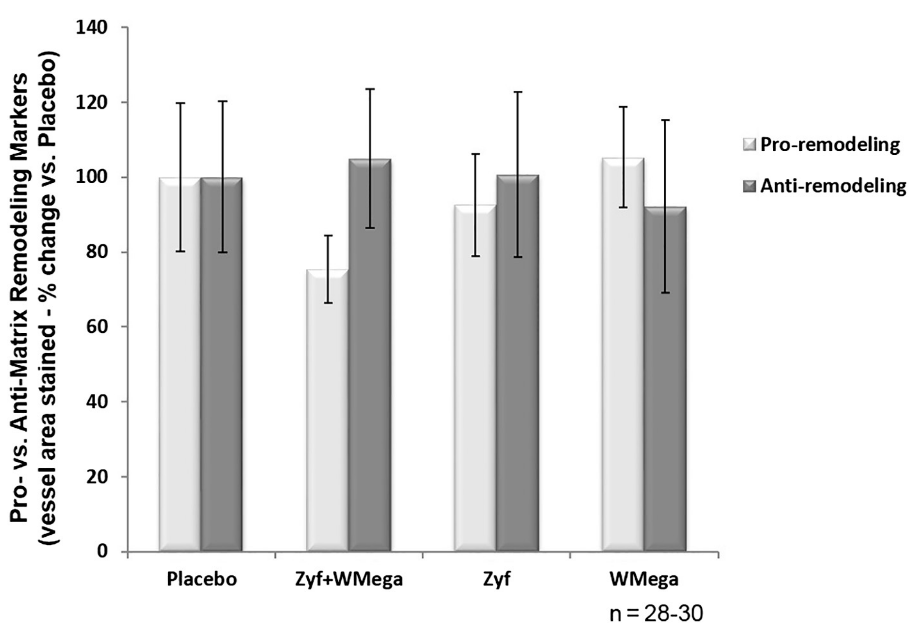

Fig. 4 – Dietary supplementation does not alter pro- or anti-remodeling isoforms. At 28d post-BA, carotid artery sections were procured and stained with MMP family isoform-specific antibodies to quantify localized vascular tissue expression. Stained % area values of all matrix enzymes and enzyme activators (MMP-2, MMP-9, MMP-14) were pooled as a measure of pro-modeling isoforms, and stained % area values of all enzyme inhibitors (TIMP-1, TIMP-2) were pooled as a measure of anti-remodeling isoforms. Abbreviations: Zyf, Zyflamend; WMega, Wholemega; Zyf + WMega, Zyflamend +Wholemega. Pooled data sets were normalized to placebo as a measure of % change, and graph exhibits means ± SEM, n = 28–30.

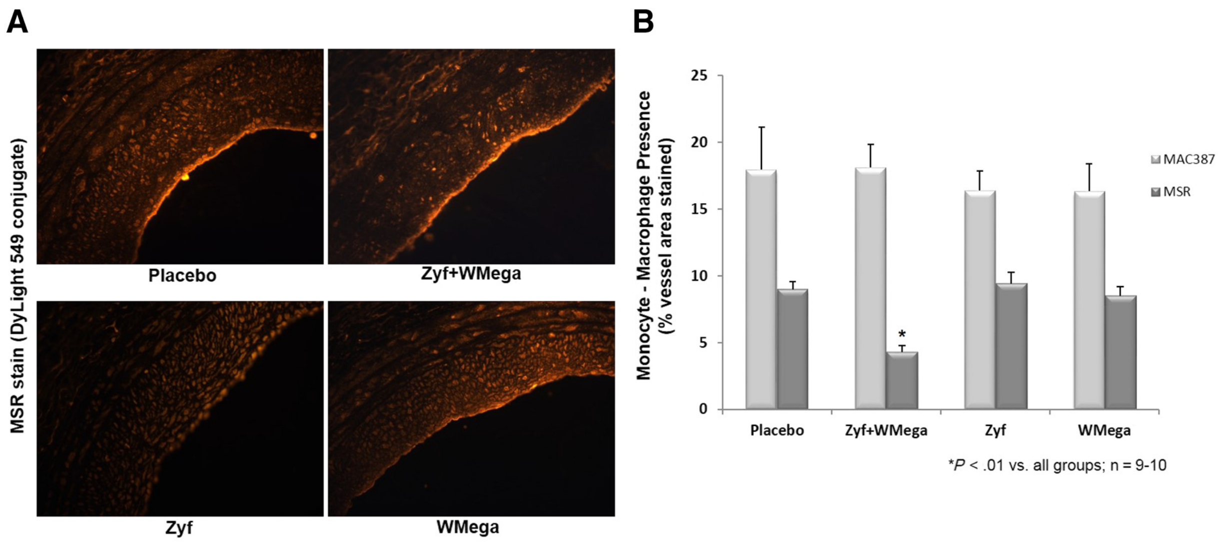

Fig. 5 – Zyf + WMega supplementation decreases MSR presence in injured arterial tissue. At 28d post-BA, carotid artery sections were procured and stained with antigen-specific antibodies for monocyte–macrophage antigen (MAC387) and M2 macrophage specific antigen macrophage scavenger receptor (MSR) to quantify cell surface marker presence. A. Images, taken at 400× magnification, are representative of stained vessel wall samples from each experimental group. Abbreviations: Zyf, Zyflamend; WMega, Wholemega; Zyf + WMega, Zyflamend +Wholemega. B. Graph exhibits means ± SEM, n = 9–10. *P<0.01 vs all other groups.

Dysfunctional remodeling of the vascular wall is considered to be a fundamental process in IH development, and phenotypic modification of medial VSMCs and their intimal migration is a major source of lesion formation. MMPs play a role in vascular remodeling due to their ability to selectively degrade components of the extracellular matrix, allowing room for this uncontrolled migration. Expression of MMPs can be influenced by pro-inflammatory cytokines, and we have demonstrated a role for unbalanced MMP regulation in pathologic vascular remodeling [14,16-18]. However, despite the reduction of IH development in Zyf + WMega animals, tissue levels of MMPs remained statistically unchanged between supplemented and placebo groups. When matrix remodeling isoforms were assayed independently, there was

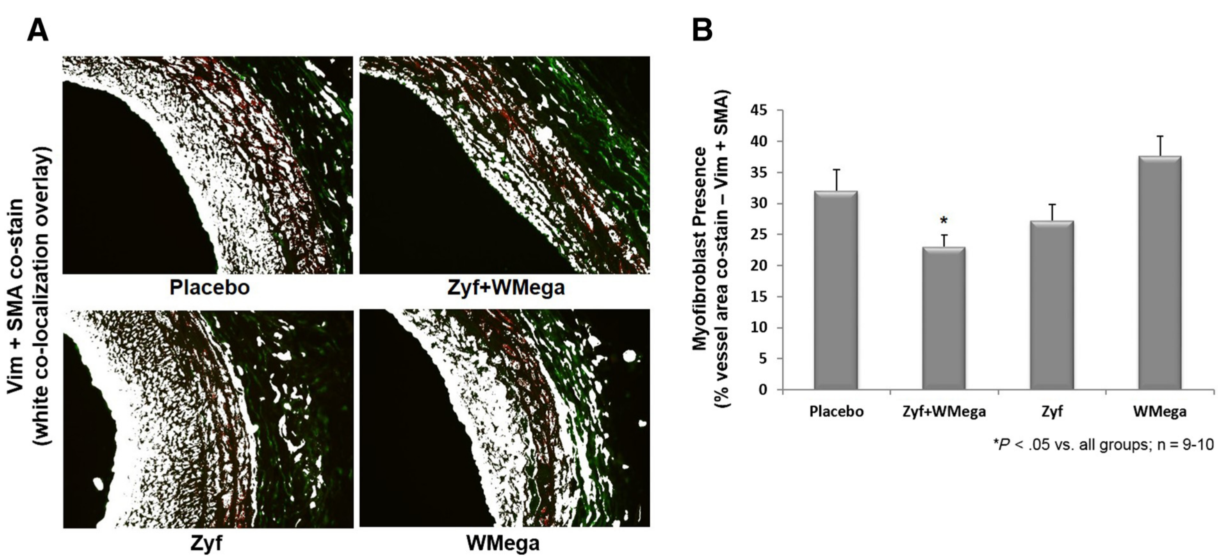

Fig. 6 – Zyf + WMega supplementation decreases Vim + SMA co-localized presence in injured arterial tissue. At 28d post-BA, carotid artery sections were procured and co-stained simultaneously with vimetin and smooth muscle α-actin (Vim + SMA) antibody cocktails to quantify co-localized cell surface marker presence indicative of M2-derived myofibroblast differentiation. A. Images, taken at 400× magnification, are representative of stained vessel wall samples from each experimental group. Green fluorescent areas are stained with Vim-specific label. Red fluorescent areas are stained with SMA-specific label. White areas are representative of Vim + SMA co-localization. B. Graph exhibits means ± SEM, n = 9–10. *P<0.05 vs all other groups.

no significant difference in the tissue expression of any isoform between experimental groups. When isoforms were categorized into pro- vs anti-remodeling classes based on their known selective mode of action, the pro-remodeling class was slightly decreased in the Zyf + WMega group, but not to a statistically significant degree. Therefore, the protective effects seen here appear independent of inflammatory-modulated matrix remodeling mechanisms known to play a role in vessel disease progression.

Recently an additional mechanism of IH development has been demonstrated. Studies have shown that monocyte/macrophage infiltration is an independent predictor of restenosis and can contribute to IH development by serving as precursors to neointimal myofibroblasts [13,29]. Moreover, monocyte infiltration has been shown to play a specific role in age-related IH development [30]. In the current study, inflammatory screening indicated macrophage chemokines IP-10 and MIP-2 were decreased in the serum of Zyf + WMega animals during long-term chronic inflammatory processes. We proposed that the downregulation of these potent chemoattractants, both secreted by and targeted to signaling of monocytes/macrophages, could be playing a role in the inhibition of macrophage infiltration to the injured vessel wall. Staining for cell surface receptor MAC387 revealed that overall macrophage/monocyte infiltration was not affected in the tissue of any group. However, staining for MSR revealed less Type 2 (M2) macrophages present in the tissue of Zyf + WMega-supplemented animals compared to placebo controls. M2 macrophages are primarily responsible for moderating the inflammatory response, eliminating cell debris, scavenging LDL cholesterol, and promoting tissue remodeling. Because M2 macrophages can serve as precursors to myofibroblast differentiation and can contribute to matrix deposition, we used Vim + SMA co-stain to determine the degree of M2-dervied myofibroblasts present in the injured vessel wall. Indeed, we demonstrated that Zyf + WMega limited the alternative differentiation of M2 macrophages and reduced the presence of M2-derived myofibroblasts. The attenuation of matrix-deposition-induced fibrosis is a possible mechanism of Zyf + WMega controlling IH development and influencing vascular function and vasotone.

Prior to our study, Zyf had not been investigated in the prevention of vascular disease progression or its pro-inflammatory modulated pathways. Here we demonstrated that dietary supplementation with human equivalent doses of Zyf + WMega significantly inhibited IH development in vivo in an animal model of vascular injury. We also show that this attenuation is an apparent result of the interaction between Zyflamend and Wholemega in combination above and beyond what was observed with each of them alone. Evidence here suggests that Zyf + WMega's vascular protective effects may be imparted by the control of M2 macrophage infiltration as precursors as neointimal myofibroblast differentiation. Improvements to the mechanistic arm of this study could be made with more deliberate cytokine examination and individual inflammatory pathway analyses. In addition, while our group has previously demonstrated synergism for the individual herbal component of Zyf [2], we have not yet established a combinatorial index for Zyflamend plus Wholemega administration in order to validate their affects as additive vs synergistic. This could be an area of future investigation. Furthermore, in vivo studies on alternations in vascular sheer stress and vasotone could further support the inhibition of myofibroblast differentiation as the means of protection. None the less, these results support our research hypothesis that Zyflamend and/or Wholemega may be beneficial for vascular protection and suggest that this unique dietary supplement combination may prove more effective than individual administration alone. Future clinical investigations are warranted to examine this combination as a potential therapeutic anti-inflammatory agent in the prevention of vascular disease progression.

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